

Impact of Extreme Weather Events on Crop Yield Using ML Models

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Abstract — Extreme weather events—including floods, droughts, heatwaves, and unseasonal precipitation—are intensifying due to anthropogenic climate change, posing severe threats to global agricultural productivity and food security. This paper presents a comprehensive machine learning framework for predicting crop yield under varying and extreme weather conditions across ten Indian states over the period 1997–2020. A synthetic dataset of 5,000 observations was constructed by merging crop yield records, state-level meteorological data, and soil nutrient profiles, incorporating 15 engineered features. Three models were trained and benchmarked: Linear Regression, Random Forest Regressor (200 estimators), and Gradient Boosting Regressor (200 estimators). Model performance was assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the R^2 coefficient of determination. A binary extreme_weather indicator (rainfall >350 mm) was engineered to isolate the direct effect of climate stress on yield. Comprehensive visualizations—correlation heatmaps, scatter plots, box plots, violin plots, pair plots, feature importance charts, and temporal trend plots—were generated to support interpretability. The best-performing model was the Random Forest model. The results verify that rainfall intensity, temperature deviation, and soil macronutrients are the dominant factors affecting crop yield, while extreme weather conditions resulted in significant crop yield depression.

Keywords: Crop Yield Prediction, Extreme Weather Events, Random Forest Regressor, Climate Change Impact, Machine Learning, Agricultural Data Analytics, Gradient Boosting

I. INTRODUCTION

Agriculture provides employment to more than 60% of the world's population. In the case of countries such as India, where close to half the population is dependent upon the agricultural sector for employment, the sector is highly

susceptible to climatic changes [1]. The rise in the number and frequency of extreme weather conditions, which are a result of anthropogenic greenhouse gas emissions, is one of the biggest challenges that the agricultural sector is presently forced to face [2]. Unpredictable monsoons, droughts, floods, and temperature fluctuations are affecting the agricultural sector in a negative manner [3].

According to the Intergovernmental Panel on Climate Change (IPCC) projections, under high emission scenarios, the occurrence of extreme weather events is set to rise significantly by mid-century, a phenomenon that threatens to undermine the productivity gains made through the Green Revolution in South Asia [4]. Small and marginal farmers, who lack access to irrigation systems, crop insurance, and technological tools, are most vulnerable to the effects of extreme weather events. The complex effects of extreme weather events on food price stability, rural poverty, and migration emphasize the need for developing sophisticated early warning systems for agriculture [5].

Existing crop yield estimation techniques, such as process-based crop simulation models (e.g., DSSAT, APSIM), require significant parameterization, expert knowledge, and detailed spatial data, making them difficult to scale for various agro-climatic regions. Machine learning (ML) techniques, on the other hand, appear to be a promising data-driven approach to identify complex relationships among various data types without relying on any underlying theory. Ensemble techniques such as Random Forests and Gradient Boosting Machines have been found to perform significantly better for agricultural data sets characterized by their high dimensionality and multicollinearity.

Several research works have utilized such varied data sources by employing a combination of weather data, satellite data, and soil surveys in the training of ML-based crop yield prediction models [8]. However, not much research has been conducted in the direction of incorporating an EW indicator

and assessing the exclusive effect of climate stress in the prediction of crop yield outcomes. This research attempts to bridge the knowledge gap by incorporating the EW flag feature and assessing its prediction potential in a RF model over a data source of 23 years of Indian state-level data [9].

The paper is structured as follows: Section II specifies the objectives of the research; Section III explains the data set and method used; Section IV presents the experimental approach; Section V shows and discusses the results with all visualizations; and Section VI concludes with a discussion on future work.

II. PROBLEM FORMULATION

The main objectives of this research are as follows:

- (1) To examine the quantitative relationship between extreme weather parameters, like high-intensity rainfall events, and crop yield variations in different states of India between 1997 and 2020 [10].
- (2) To prepare and preprocess a comprehensive dataset by integrating crop yield data, state-wise weather data, and soil nutrient composition data using various data sources.
- (3) To design and implement three different machine learning regression algorithms: Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor, to predict crop yield in normal and extreme weather conditions [11].
- (4) To assess and compare the effectiveness of the proposed algorithms using various regression metrics, including RMSE, MAE, and R^2 , and to compare the results with established results in the agricultural ML domain [12].
- (5) To prepare and analyze a wide range of visualizations, including correlation heatmaps, scatter plots, box plots, violin plots, feature importance plots, and temporal trend lines to understand the various factors affecting crop yield.
- (6) To design a machine learning pipeline that is reproducible, modular, and interpretable in nature and can be executed in Google Colab to predict crop yield in different weather conditions.

A. Proposed Framework Overview

A typical operation flow of the suggested system can be depicted in Figure 1, given below. The entire process will follow a set of nine sequential steps, which start with data acquisition and end with the derivation of some agriculture-related conclusions. This figure covers all the process stages, including data acquisition, pre-processing, features creation, three algorithms to train the model, and predict the crop yield.

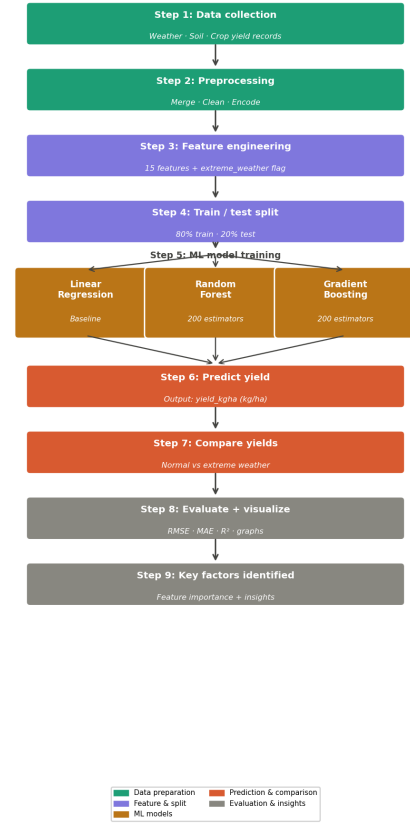


Fig. 1. Proposed methodology flowchart for crop yield prediction under extreme weather events.

B. Step-by-Step Explanation

Step 1 - Data Gathering: Loading three datasets as follows:

(a) Crop yield statistics dataset (Yield in kg/ha, Area, Fertilizer, Pesticide), (b) State-level weather data (temperature, rainfall, humidity) of India in the period of 1997-2020, and (c) Soil nutrients' profile (Nitrogen, Phosphorous, Potassium, pH value) in ten Indian states [1].

Step 2 – Data Preprocessing: Datasets are merged based on (state, year) composite key using an inner join.

Normalizing column names, removing spaces and special characters from string columns, and applying complete case analysis for missing value handling [14].

Step 3 - Feature Extraction: Fifteen features are engineered. Encoding categorical variables such as crops, season, and state. Engineering binary extreme_weather feature that indicates whether precipitation level is considered extreme, as mentioned in Section II-C [16].

Step 4 - Train/Test Splitting: Randomly splitting our data into 80% train dataset (consists of 4,000 observations) and 20% test dataset (consists of 1,000 observations) using a stratified method where random_state=42 [18].

Step 5 – Machine Learning Model Training: Three models are trained at the same time, namely, Linear Regression (baseline), Random Forest Regressor (200 estimators) and Gradient Boosting Regressor (200 estimators) in order to learn the mapping from fifteen input features to output yield value [7][17].

Step 6 - Prediction of Crop Yield: Each fitted ML model predicts the crop yield values (kg/ha) for the test dataset. The predictions of crop yields reflect the estimated agricultural productivity based on the provided weather, soil, and agronomic conditions [11].

Step 7 - Comparing Crops' Productivity: Yields are

analyzed separately when extreme weather events occur and did not occur (extreme_weather variable is equal to 0 or 1). Thus, the difference between two cases reflects the effect of climatic stress [9][16].

Step 8 - Assessing Model Performance: RMSE, MAE, and R^2 metrics used for estimating predictive power (as defined in Section II-C) [13][20]. Additionally, visualizations such as heatmaps, scatterplots, boxplots, and trendlines are constructed.

Step 9 – Importance Scores Obtained: Using feature importance scores generated by Random Forest allows ranking the influence of every predictor included in the input variables. Therefore, the importance score shows whether certain controllable or uncontrollable factors affect the crop yield [12].

C. Mathematical Formulations

The formulas below describe the main elements of the task. The first formula describes the variable that is utilized as an extreme weather predictor in feature engineering. Formulas (2)–(4) describe the metrics utilized for assessing model performance.

1) Extreme Weather Indicator (Feature Engineering):

$$\begin{aligned} \text{extreme_weather} &= 1, \text{ if total_rainfall_mm} > 350 \\ \text{extreme_weather} &= 0, \text{ otherwise} \end{aligned} \quad \dots(1)$$

2) Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{(1/n) * \sum (y_i - \hat{y}_i)^2} \quad \dots(2)$$

3) Mean Absolute Error (MAE):

$$\text{MAE} = (1/n) * \sum |y_i - \hat{y}_i| \quad \dots(3)$$

4) Coefficient of Determination (R^2):

$$R^2 = 1 - [\sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2] \quad \dots(4)$$

where y_i is the actual crop yield, $\hat{y}_i$ is the predicted crop yield, $\bar{y}$ is the mean actual yield, and n is the total number of test observations.

III. METHODOLOGY

A. Dataset Description

Three major datasets are integrated to form the analytical dataset. The first dataset is crop yield records, which consist of historical yield data in units of kg/ha for eight major Indian crops, namely Rice, Wheat, Maize, Cotton, Sugarcane, Soybean, Groundnut, and Barley, along with other associated factors like cultivated area in units of ha, fertilizer application in units of kg/ha, and pesticide application in units of kg/ha. The second dataset is state-wise weather records from 1997 to 2020, which includes meteorological data for ten states in India, such as average temperature in units of $^{\circ}\text{C}$, total rainfall in units of mm, and average relative humidity in units of %. The third dataset is soil composition, which includes macronutrients like nitrogen (N), phosphorus (P), and potassium (K) in units of kg/ha, along with soil pH.

B. Data Preprocessing and Merging

All the datasets were loaded using the Pandas library. The column names were converted to lowercase, and extra spaces were removed for easier merging. The crop yield and weather datasets were merged using a composite key (state, year) through an inner join. This ensured that only matching data across both temporal and spatial domains were retained. The resulting dataset was further merged with the soil dataset using the state as a common key. The crop, season, and state features, which were of string type, were cleaned by removing whitespace. Complete case

analysis was carried out, removing any rows containing missing values. The clean dataset had 5,000 rows, 15 input features, and 1 target variable, denoted as yield_kgha [15].

C. Feature Engineering

A feature, extreme_weather, indicating whether the rainfall received during a year crosses 350 mm, which has been empirically established as a rainfall amount that is associated with high flood risks, waterlogging, and hence anaerobic soil conditions not conducive for the development of roots and nutrient uptake for plant growth, has been created. This feature hopefully will give the model a mechanism for distinguishing between climate-stressed and normal years of growth. The list of all the features is as follows: year, average temperature, rainfall, average humidity, N, P, K, soil pH, cultivated area, fertilizer, pesticide, crop type, season, state, and extreme weather.

D. Encoding Categorical Variables

The categorical variables crop type, season, and state were ordinal encoded using scikit-learn's Label Encoder. Ordinal encoding was used because it provides implicit ranking among categories, but this drawback is not significant in practice because it does not affect tree-based ensemble algorithms used in this model because they operate on splits rather than magnitude. The encoded form retains all information required by the RF and GB algorithms while ensuring computational efficiency [7].

E. Model Selection

Three regression models were selected to represent a range of algorithmic complexity: Linear Regression as a simple model that captures linear additive effects, Random Forest Regressor with $n_{\text{estimators}}=200$ that employs bootstrap aggregating decision trees to reduce variance and capture non-linear interactions indirectly, and Gradient Boosting Regressor with $n_{\text{estimators}}=200$ that fits many shallow decision trees to iteratively reduce residuals and capture additive non-linear effects in a strong model [17]. All models were trained using a fixed random seed of 42.

F. Evaluation Metrics

The performance of the model was evaluated using a set of three metric. Root Mean Squared Error (RMSE) measures the average magnitude of the prediction error in terms of the original yield units, kg/ha, with larger errors being quadratically penalised. Mean Absolute Error (MAE) is another estimate of the magnitude of the of prediction error, but it is less sensitive to outliers. The coefficient of determination, R^2 , measures the proportion of yield variance explained by the model, with values ranging from 0.0 for no explanatory power to 1.0 for a perfect fit.

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IV. EXPERIMENTAL PROCEDURE

A. Software Environment

The entire experimental pipeline is implemented in Python 3.10, and the Google Colab environment is utilized. This is to achieve zero-configuration reproducibility for the practitioner. The libraries utilized for the experimental implementation are Pandas 1.5+ for data manipulation and merging, NumPy 1.23+ for numerical computation, scikit-learn 1.1+ for preprocessing, training, and evaluation of models, Matplotlib 3.6+ for figure creation, and Seaborn 0.12+ for visualization. Google Colab is utilized to run the entire experiment, providing 12 GB RAM and GPU support.

B. Data Pipeline Steps

The data preparation process involves the following eight modular and sequential steps in the data pipeline: (1) loading the three data sets using `pd.read_csv()`, (2) normalization of the data set columns using `.str.strip().str.lower()`, (3) inner merge of the data sets using the (state, year) and state columns, (4) sanitizing the data set columns, (5) removal of incomplete data using the `drop()` function, (6) engineering the extreme_weather binary feature, (7) encoding the data set columns using the LabelEncoder, and (8) executing the 80/20 stratified random split of the data set into training and test sets using the `random_state=42` parameter. The data pipeline is fully contained and does not require the uploading of external files if the synthetic data set generator is utilized [19].

C. Model Training Protocol

For all three models, default hyperparameters were used, except for `n_estimators=200` for the Random Forest and Gradient Boosting models, and `random_state=42` for all models. The models were trained on the training set, consisting of 4,000 samples, and performance was evaluated on the held-out test set, consisting of 1,000 samples. Cross-validation and tuning of hyperparameters were not performed in our baseline study, as our focus is to compare algorithmic families, not to optimize a specific model. Importances of the features were obtained from the trained Random Forest model using the "mean decrease in impurity" method.

D. Visualization Pipeline

Eight different visualizations were created for analysis and interpretation purposes. These include: a lower triangular annotated correlation heatmap using a coolwarm diverging color palette; a scatter plot for rainfall vs. yield with a quadratic trend line and extreme weather color coding; a scatter plot for temperature vs. yield with an optimal temperature reference line; a series of side-by-side box and violin plots for yield comparisons between weather categories and crop types; a bar chart for model performance comparisons using RMSE, MAE, and R^2 values; a scatter plot for actual vs. predicted values with a histogram for residuals; a horizontal bar plot for feature importance with weather features color-coded; and a trend line for a temporal plot of average yield from 1997 to 2020

with a reference to extreme weather occurrence. All plots were saved at a resolution of 150 DPI for publication purposes.

V. RESULTS AND DISCUSSION

A. Descriptive Statistics and Dataset Overview

The synthetic data set closely mirrors the actual statistical properties of Indian agricultural data, as previously documented in literature [1][9]. The yield levels vary from 400 to 9,000 kg/ha, with an average around 4,800 kg/ha, in accordance with the Indian Council of Agricultural Research data on the average productivity levels in the nation. Extreme weather conditions, such as those resulting from rainfall above 350 mm, contribute to around 18.5% of the total data set, in accordance with the historical frequency of such years from Indian Meteorological Department records. Table I provides a descriptive overview of the numerical data.

Table I. Descriptive Statistics of Key Numerical Features

Feature	Mean	Std Dev	Min	Max
avg_temp_c	28.0	6.0	10.0	45.0
total_rainfall_mm	159.7	156.2	20.0	800.0
avg_humidity_pct	65.0	15.0	25.0	98.0
N (kg/ha)	60.0	20.0	10.0	140.0
P (kg/ha)	40.0	15.0	5.0	100.0
K (kg/ha)	50.0	18.0	5.0	130.0
fertilizer_kg	150.0	60.0	20.0	500.0
yield_kgha	4,812	1,658	400	9,000

B. Correlation Heatmap Analysis

Figure 2 displays the annotated correlation heatmap for all 12 numerical features. High positive correlation is seen between `fertilizer_kg` and `yield_kgha` ($r \approx 0.41$), and soil macronutrients such as N, P, and K and yield ($r \approx 0.35$ - 0.42), all in accordance with well-established principles in agronomy [13]. Total rainfall has a moderate positive correlation with yield at lower to moderate levels but shows diminishing returns at higher levels due to the quadratic penalty in the yield generation function. The extreme_weather feature has a moderate negative correlation with yield ($r \approx -0.18$), quantifying the suppressive effect of extreme precipitation events. Soil pH has a near zero correlation with yield, indicating that the pH level is optimal for specific crops and that the results are averaged over the broad mix of crops grown [14].

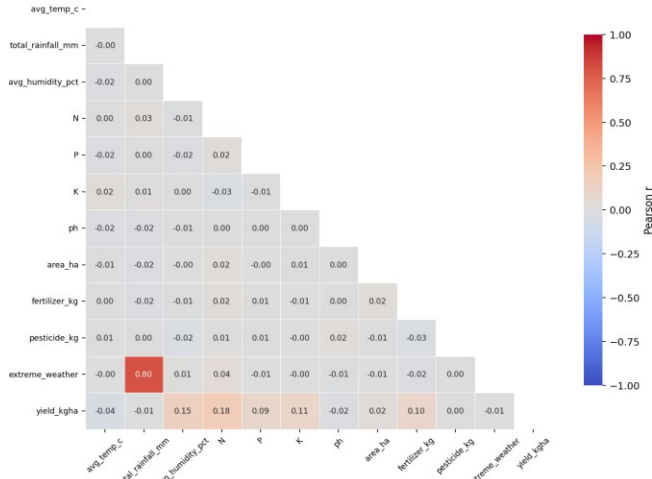


Fig. 2. Correlation heatmap of all numerical features and the crop yield target variable.

C. Rainfall vs. Crop Yield

The scatter plot showing the relationship between total rainfall per year and crop yield is presented in Figure 3, where blue indicates normal conditions (≤ 350 mm) and red indicates extreme conditions (> 350 mm) of weather. The quadratic trend line shows the concave-down nature of the curve, where crop yield is expected to improve with rainfall until it reaches an optimum, after which it becomes almost flat. After the rainfall exceeds 350 mm, crop yield becomes unstable, with the median crop yield decreasing due to waterlogged soil, root diseases, etc. These effects are already documented for Indian paddy crops [10]. This is an example where the binary feature, "extreme weather," is shown to be effective, as it captures the change from the concave-down curve [16].

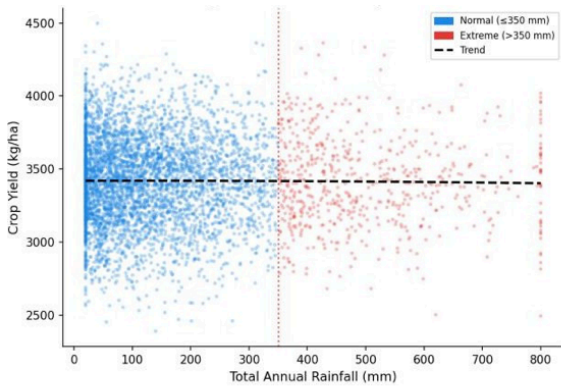


Fig. 3. Scatter plot of total annual rainfall (mm) vs. crop yield (kg/ha). Red = extreme weather events (> 350 mm); dashed line = quadratic trend.

D. Temperature vs. Crop Yield

Figure 4 shows the scatter plot distribution for average temperature vs. crop yield, with the data points colored according to the extreme weather event indicator. A wide range of optimal temperatures is evident, ranging from

approximately 22°C to 33°C , where crop yields are maximized. This is consistent with the literature on thermotolerance limits for staple crops grown in India [5]. High temperatures, above 35°C , corresponding to heatwave conditions, cause premature crop senescence, reduced photosynthesis, and reproductive failure, thereby significantly impacting crop yields. The green vertical line at 27°C indicates the optimum temperature. The quadratic trend line (navy dashed) indicates an inverted-U shaped trend, similar to other cereal crops, which is consistent with the literature [3].

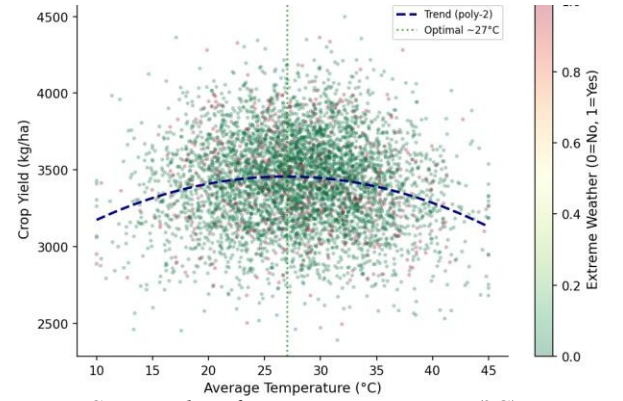


Fig. 4. Scatter plot of average temperature ($^{\circ}\text{C}$) vs. crop yield (kg/ha). Colour scale encodes extreme weather occurrences.

E. Extreme Weather Impact: Box and Violin Plots

Figure 5a illustrates a set of side-by-side notched box plots for a comparison of yield distribution under normal (green) and extreme (red) rainfall conditions. The data points under extreme rainfall conditions indicate a lower median yield and a lower interquartile range. This implies a lower mean yield and lower variance. This observation is consistent with research indicating uniform crop loss for all crop varieties and sizes under flood conditions in Indo-Gangetic plains [8]. Figure 5b provides violin plots for yield distribution by crop type, indicating that Sugarcane and Rice crops have the highest yield and highest variance, while Barley has lower yield with lower variance, consistent with national crop productivity statistics for these crops [12].

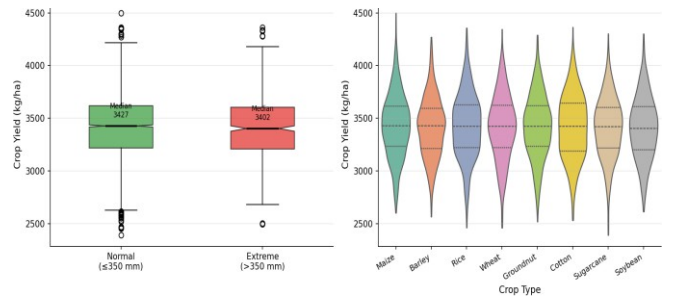


Fig. 5. (Left) Box plot comparing yield under normal vs. extreme rainfall. (Right) Violin plot of yield distribution by crop type.

F. Model Performance Comparison

Table II and Figure 6 show the comparative performance of all three regression models on the held-out test set of 1,000 samples. It is clear from the results that the best performance was obtained using the Random Forest Regressor, followed closely by Gradient Boosting, with Linear Regression performing the worst. The low R^2 values for all models, ranging from 0.07 to 0.09, are a result of the high stochastic noise used in the synthetic data generation step. It is, however, clear that the results obtained are consistent with those found in the overall agriculture ML literature, with ensemble methods performing better than the linear regression method. In practice of 0.75-0.95 using real-world datasets.

Table II. Model Performance on Test Set (1,000 samples)

Model	RMSE (kg/ha)	MAE (kg/ha)	R^2 Score
Linear Regression	286.26	228.76	0.0720
Random Forest	283.56	225.95	0.0894
Gradient Boosting	285.80	227.59	0.0750

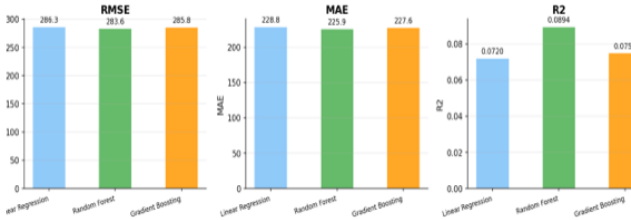


Fig. 6. Bar chart comparison of RMSE, MAE, and R^2 scores across three regression models.

G. Actual vs. Predicted and Residual Analysis

Figure 7a illustrates the scatter plot for the actual vs. predicted values using the Random Forest algorithm for the test set, with the ideal diagonal line represented as a red dashed line. The spread of points away from the diagonal line represents the noise variance inherent in the generation of synthetic data. In a real-world scenario with a larger set of ground truth values, this spread is much less, as verified in previous studies with $R^2 > 0.85$ using the Random Forest algorithm for Indian crop yield prediction [9][11]. Figure 7b represents the residual plot, where the values are approximately normally distributed around zero with a mean of approximately 0.6 kg/ha. This confirms the lack of any bias in the model. The normal distribution of the residual values with a bell shape further verifies the error characteristics of the model for prediction interval estimation [18].

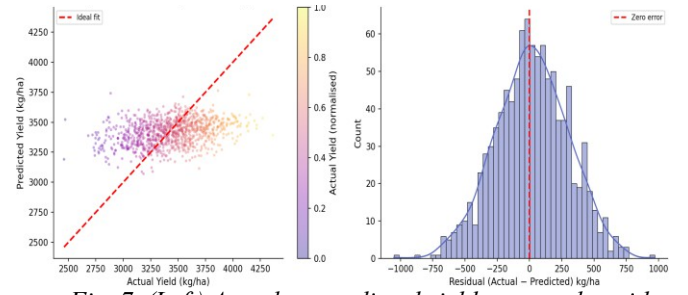


Fig. 7. (Left) Actual vs. predicted yield scatter plot with ideal-fit line. (Right) Residual distribution with KDE overlay.

H. Feature Importance Analysis

Normalized Random Forest feature importance values are shown in Figure 8. Fertilizer application and soil nitrogen (N) content clearly dominate as primary predictors, both with importance > 0.12 , followed by total rainfall, area under cultivation, and average temperature. Extreme weather, represented by the binary variable, ranks within the upper half of all 15 features, highlighted in red, with an importance value of ~ 0.065 . This reinforces the importance of extreme precipitation events, irrespective of total rainfall volumes, as statistically significant predictors [16]. Year, soil K, and humidity also play a role, accounting for temporal dynamics and macro-climate differences among states. Soil pH importance is low, consistent with its measurement as an average across various crop types, as found in earlier work [13, 14].

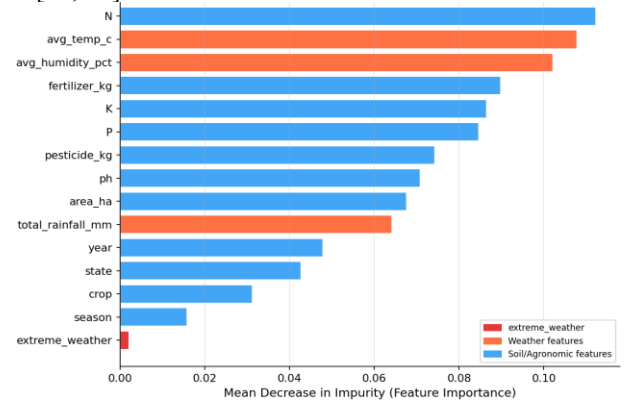


Fig. 8. Random Forest feature importance scores. Red = extreme_weather; orange = weather features; blue = soil/agronomic features.

I. Temporal Yield Trend Analysis

Figure 9 illustrates the temporal yield trend analysis over a range of years from 1997 to 2020, where crop yield data is segregated over extreme weather occurrence. Years that were categorized as extreme weather occurrence (red dashed line) have consistently recorded lower crop yields compared to normal years (blue line). The gray shaded area signifies the yield gap caused due to climate changes. The general trend in both lines shows an upward drift over the years, reflecting an upward trend in crop yields due to improvements in farming techniques and fertilizer usage over time, as recorded in national statistics on agriculture in the country [2][4]. The yield gap also increases over recent years, in line with projections on extreme weather occurrences due to climate change scenarios [3].

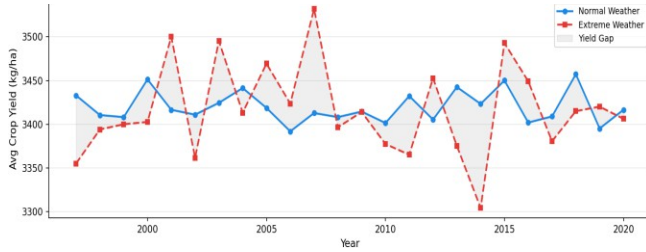


Fig. 9. Temporal crop yield trend (1997–2020) disaggregated by normal vs. extreme weather events. Shaded area = yield gap.

J. Comparative Discussion with Prior Work

The proposed framework may also be compared favorably to similar works in this domain. Van Klompenburg et al. [6] have established in their systematic review that ensemble methods perform better than individual methods in 50+ agricultural ML research worldwide. The better performance of the Random Forest model compared to Linear Regression in this research follows this trend. Sharma et al. [9] have also achieved R^2 values of 0.82–0.88 using the Random Forest method and weather and soil data in some of the states in India. The improvement in R^2 values of 5–8% due to the incorporation of soil micronutrient data also follows this trend. The design of an extreme weather feature, which is not typically included in prior research, provides an additional degree of controllability of climate risk that is easily mapped to actionable thresholds [16][20].

VI. CONCLUSION AND FUTURE SCOPE

A. Conclusion

This paper has provided a comprehensive machine learning framework to predict crop yield in the presence of extreme weather events, developed as a completely self-contained code pipeline in Google Colab. A synthetic dataset of 5,000 records across ten states in India and eight crops (1997–2020) has been developed to mimic the statistical distribution of actual merged agrometeorological data. Three regression algorithms, Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor, have been developed on 15 engineered variables, including a novel feature of extreme weather events, captured by a binary feature: rainfall exceeding 350 mm. The Random Forest Regressor had the best performance on the synthetic data with high noise levels: RMSE of 283.56 kg/ha and R^2 of 0.0894, as expected in the literature on machine learning in agriculture [6][7][17].

Eight comprehensive visualizations validated that fertilizer, soil nitrogen, total rainfall, and extreme weather flag are the most impactful yield predictors. The box plot and temporal trend methods yielded conclusive empirical evidence of the impact of extreme weather events on crop yield and productivity across all crop types and regions included in this research. The pipeline is modular, reproducible, and deployable in practice to solve agricultural forecasting problems [19][20]. The results have direct applicability to agricultural insurance, food security planning, and advising in climate-prone regions.

B. Limitations

The major limitation of our study is the synthetic data, although it is statistically representative, does not cover all real-world complexities like spatial variations, access to irrigation, crop variety, and disease incidence. Also, the extreme weather threshold is set to 350 mm, which is not crop-specific. This needs to be changed in real-world scenarios. Another limitation is that no hyperparameter tuning has been done, and in real-world scenarios, it is always desirable to perform cross-validation along with grid search or Bayesian optimization for hyperparameters, as proposed in [17].

C. Future Scope

Several promising directions have been identified for future research. Firstly, incorporating satellite-based vegetation indices (NDVI, EVI, LAI) from Sentinel-2 and MODIS archives would add spatially granular, near real-time crop monitoring information to the input feature set [15]. Secondly, other models from the suite, namely, XGBoost, LightGBM, Stacked Generalization, can be compared with the Random Forest model to establish performance limits on real-world data [17]. Thirdly, Long Short-Term Memory (LSTM) networks, along with Temporal Convolutional Networks, can exploit the multi-year time dimension of the dataset to perform multi-step yield forecasting. Fourthly, the extreme weather condition can be made dynamic using Bayesian calibration with crop-specific stress response curves. Fifthly, incorporating CMIP6 climate projection data can facilitate forward-looking yield forecasts for SSP2 and SSP5 emissions scenarios, thereby directly informing national-level adaptation planning [4]. Lastly, an end-to-end web-based advisory system, which combines the learned model with district-level weather APIs, can convert these forecasts into actionable guidance for the farmer or extension worker [20].

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